PADMAVATHI M

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Profile Summary

- Software Developer proficient in Java, Python, and JavaScript, with a strong focus on building scalable web applications and reliable backend solutions.
- Hands-on experience with React.js, Node.js, Express.js, and REST APIs, developing responsive and data-driven applications with practical industry experience.
- Experienced in exploring RAG, semantic search, and machine learning techniques to develop intelligent and practical solutions for real-world challenges.

Education

B.E. Computer Science and Engineering 2023 – 2027
Saveetha Engineering College, Chennai **CGPA: 8.62**

Skills

Programming Languages: Java, Python, JavaScript
Frontend Development: React.js, Tailwind CSS, HTML5, CSS3
Backend Development: Node.js, Express.js, FastAPI, Flask, RESTful API Design
Databases & Cloud: MySQL, MongoDB, Firebase
AI Concepts: RAG, Prompt Engineering, Machine Learning, Semantic Search
Developer Tools: Git, GitHub, Postman, Jupyter Notebook, Streamlit

Experience

MERN Stack Intern — The Hindu Group, Chennai Aug 2026 – Oct 2026

- Developing full-stack web applications using MongoDB, Express.js, React.js, and Node.js, following modular and scalable development practices.
- Exploring semantic search and AI-powered technologies, working with text chunking, embeddings, vector search, and Retrieval-Augmented Generation (RAG).

Web Developer Intern — Geo Pacific Solutions, Chennai Jan 2026 – Mar 2026

- Developed web application features using React.js and Spring Boot, and built RESTful APIs to connect frontend components with backend services.

Projects

EyeQ-DR-G — AI-Powered Retinal Disease Detection **GitHub**

- Developed an AI-powered retinal disease detection system using ResNet50 to classify fundus images for diabetic retinopathy and glaucoma.
- Built a Flask-based backend and React.js frontend to enable image-based prediction, with Grad-CAM heatmaps for visualizing the regions influencing model predictions.
- **Result:** Delivered a highly accurate system (91%) automating and streamlining diagnostic reporting workflows.

Task Management System — Scalable Workflow Automation **GitHub**

- Architected a comprehensive task tracking software using JavaScript, optimizing client-side performance.
- Implemented core logic for task delegation, real-time status tracking, and automated workflow triggers.
- **Result:** Reduced manual tracking overhead by providing a responsive user interface and robust data handling.

Social Media Aggregator — Asynchronous Data Pipeline **GitHub**

- Developed a centralized aggregation platform utilizing advanced asynchronous programming in JavaScript to handle concurrent API requests.
- Designed a high-performance UI to render aggregated data feeds dynamically without blocking the main thread.
- **Result:** Improved data retrieval efficiency and created a unified platform for cross-network engagement.

Achievements & Certifications

- **Hackathon Veteran:** Actively participated in **40+ hackathons**, driving rapid prototyping under pressure.
- **Hackathon Placements: Winner** at QTuxathon 3.0 Hackathon | **3rd Runner-Up** at Impacteers Hackathon.
- **Top 40 Finalist** (out of 2,000+ competitors) — StartupTN Grand Finale.
- **Certifications:** Google Data Analytics Professional Certificate (2026) | Machine Learning Certification (2024) | Front End Web Developer Certification (Infosys, 2025).